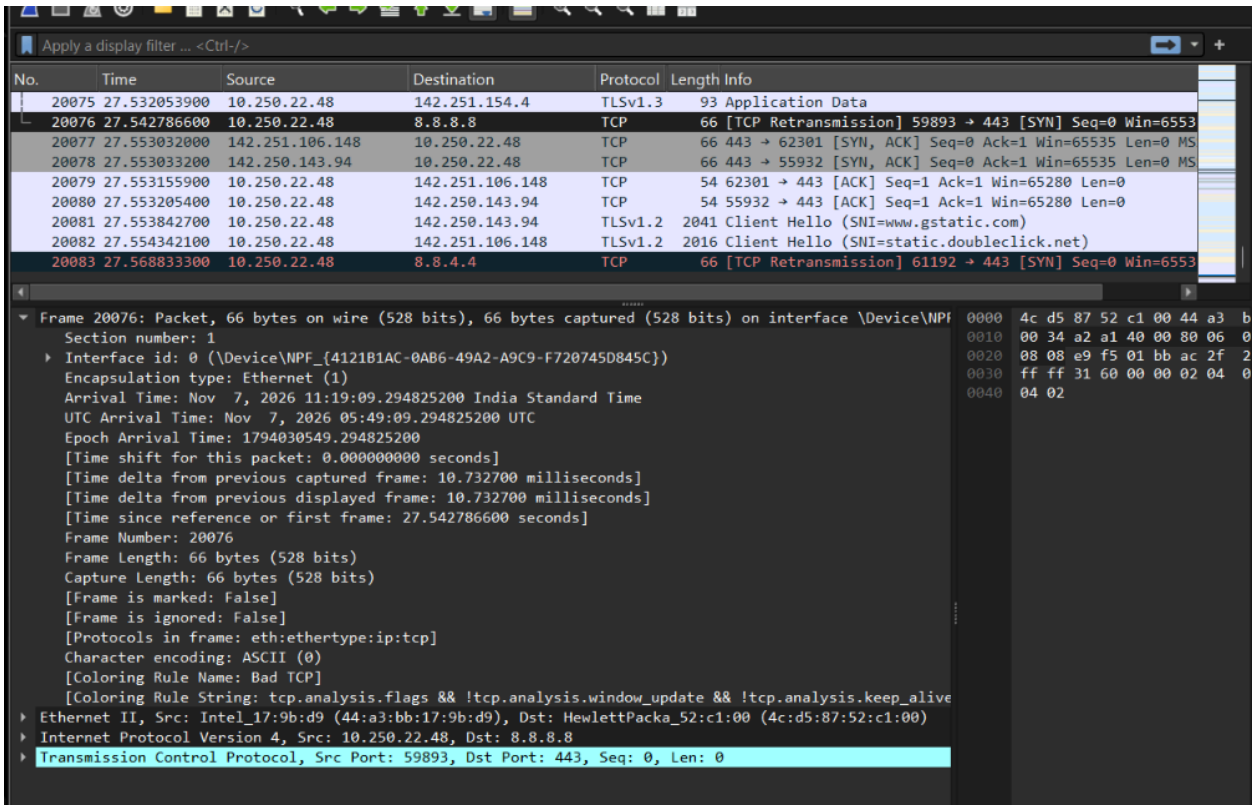
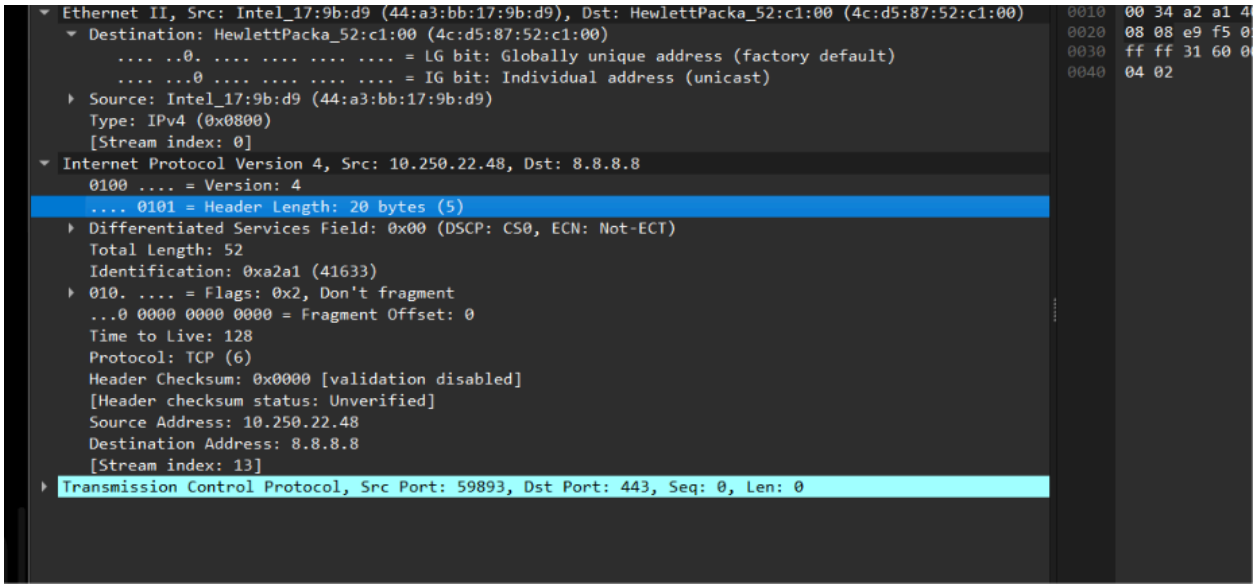
CNNP SKILL-1

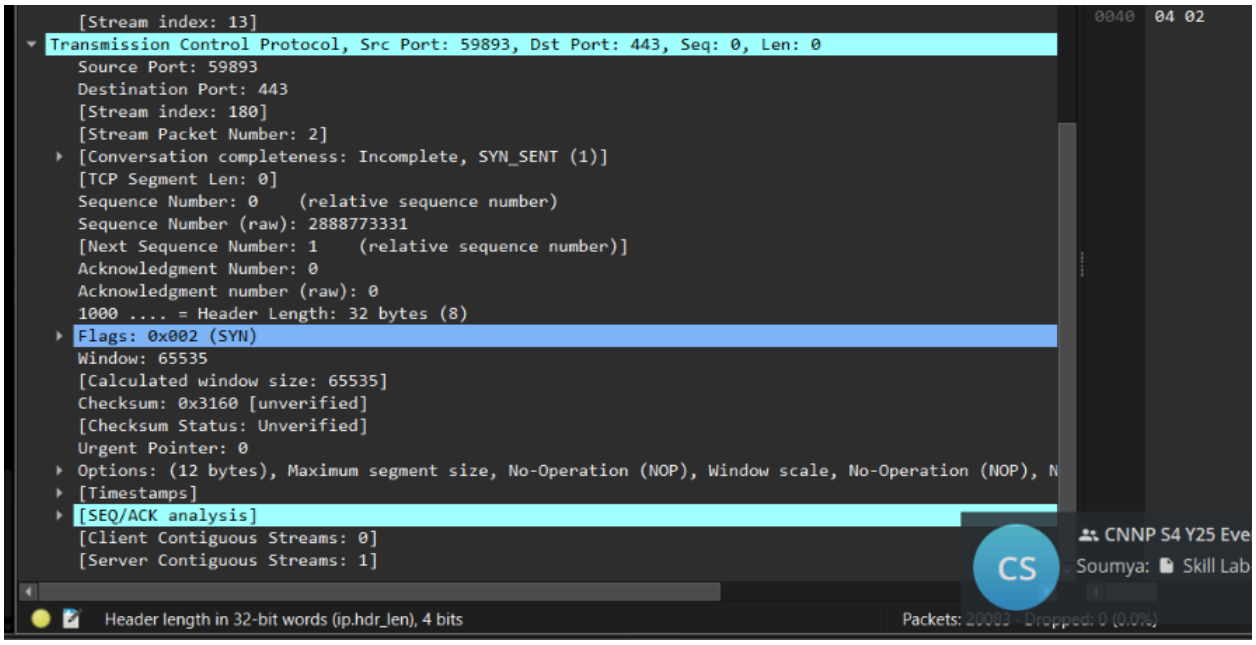
Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission. Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead.

Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data.




Ethernet II, Src: Intel_17:9b:d9 (44:a3:bb:17:9b:d9), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Source: Intel_17:9b:d9 (44:a3:bb:17:9b:d9)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 10.250.22.48, Dst: 8.8.8.8
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 52
Identification: 0xa2a1 (41633)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.22.48
Destination Address: 8.8.8.8
[Stream index: 13]
Transmission Control Protocol, Src Port: 59893, Dst Port: 443, Seq: 0, Len: 0
0010 00 34 a2 a1 4
0020 08 08 e9 f5 0
0030 ff ff 31 60 0
0040 04 02


[Stream index: 13]
Transmission Control Protocol, Src Port: 59893, Dst Port: 443, Seq: 0, Len: 0
Source Port: 59893
Destination Port: 443
[Stream index: 180]
[Stream Packet Number: 2]
[Conversation completeness: Incomplete, SYN_SENT (1)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 2888773331
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1000 .... = Header Length: 32 bytes (8)
Flags: 0x002 (SYN)
Window: 65535
[Calculated window size: 65535]
Checksum: 0x3160 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), N
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 0]
[Server Contiguous Streams: 1]
0040 04 02
CS
CNNP S4 Y25 Eve
Soumya: Skill Lab
Header length in 32-bit words (ip.hdr_len), 4 bits
Packets: 20093 · Dropped: 0 (0.0%)

Parameter	TCP Packet (bytes)	UDP Packet (bytes)
Frame length	66	78
Ethernet header length	14	14
Ip header length	20	20
Transport header length	32	8
Ip total length	52	64
Transport header length	32	8
payload	0	56